

DESIGN AND ANALYSIS OF ALGORITHMS

PRACTICAL REPORT – PRACTICAL NO. 9

Practical No. 9	Experiment Bellman-Ford Algorithm
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1. Aim

To find shortest paths from a single source in a weighted graph using Bellman-Ford and detect negative-weight cycles.

2. Objective

- Understand repeated edge relaxation.
- Find shortest paths when negative edge weights are present.
- Detect a reachable negative-weight cycle.

3. Theory

Bellman-Ford is a single-source shortest path algorithm that supports negative edge weights. Unlike algorithms that require non-negative edge weights, Bellman-Ford repeatedly relaxes every edge to progressively improve the shortest known distance from the source. For a graph with V vertices, all edges are relaxed $V - 1$ times. This ensures that shortest path information can propagate through the graph even when negative edge weights are present. After these iterations, one additional pass is performed. If any distance can still be reduced, the graph contains a reachable negative-weight cycle. The time complexity of the algorithm is $O(VE)$.

4. Working / Procedure

First, initialize the distance of the source vertex to 0 and the distance of every other vertex to infinity. Next, examine every edge and perform relaxation: if the distance to the destination through the current edge is smaller than its currently stored value, update the distance. Repeat this process for $V - 1$ iterations. Finally, examine all edges once more. If any distance can still be reduced, a reachable negative-weight cycle is present. Otherwise, the final distance values represent the shortest paths from the selected source vertex.

5. Conclusion

Bellman-Ford computes shortest paths in graphs containing negative edge weights and can detect reachable negative-weight cycles. The practical demonstrates the importance of repeated relaxation for obtaining correct shortest-path distances.